## Experiment 1 – Linear Regression

**1. $R^2$**

$$SSE = \sum (y_i - \hat{y}_i)^2$$

$$SST = \sum (y_i - \bar{y})^2$$

$$R^2 = 1 - \frac{SSE}{SST}$$

**2. Adjusted $R^2$**

$$R^2_{adj} = 1 - (1 - R^2)\frac{n-1}{n-p-1}$$

**3. t-statistic**

$$t = \frac{\beta}{SE(\beta)}$$

**4. 95% Confidence Interval**

$$CI = \beta \pm 1.96(SE)$$

**5. Logistic Regression – Odds Ratio**

| Feature | Coefficient | Odds Ratio | Percentage change |
|---|---|---|---|
| Age | 0.06 | 1.06 | +6% |
| Income | 0.85 | 2.34 | +134% |
| Loan Amount | -0.40 | 0.67 | -33% |
| Credit Score | 1.25 | 3.49 | +249% |

## Experiment 2 – Decision Tree and KNN

**1. Gini Impurity**

Dataset: 6 Yes, 4 No.

$$Gini = 1 - (0.6)^2 - (0.4)^2$$

$$= 1 - 0.36 - 0.16 = \boxed{0.48}$$

**2. Gini after split**

Left: 4 Yes, 0 No.

$$Gini_L = 0$$

Right: 2 Yes, 4 No.

$$Gini_R = 1 - (2/6)^2 - (4/6)^2 = 0.4444$$

Weighted Gini:

$$= \frac{4}{10}(0) + \frac{6}{10}(0.4444)$$

$$= \boxed{0.2667}$$

**3. Gini Gain**

$$= 0.48 - 0.2667 = \boxed{0.2133}$$

**4. KNN Classification**

Query: $Q = (3, 3)$

| Point | Coordinates | Distance |
|---|---|---|
| A | (1,1) | 2.828 |
| B | (2,2) | 1.414 |
| C | (4,4) | 1.414 |
| D | (5,5) | 2.828 |

For $K = 3$, nearest neighbours: B, C, A (or D, depending on tie handling).

If their classes are B = Yes, C = Yes, A = No:

$$\boxed{\text{Prediction} = \text{Yes}}$$

**5. KNN Regression**

Nearest neighbour values: 20, 30, 40.

$$Prediction = \frac{20 + 30 + 40}{3}$$

$$= \boxed{30}$$

## Experiment 3 – Approximate Shapley Values

Given:

$$f(x) = 2x_1 + 3x_2 + 5x_3$$

$$x = (4, 2, 3), \quad z = (5, 1, 4)$$

Initial prediction:

$$f(z) = 2(5) + 3(1) + 5(4) = 33$$

Final prediction:

$$f(x) = 2(4) + 3(2) + 5(3) = 29$$

Feature contributions:

$$\phi_1 = 2(4 - 5) = \boxed{-2}$$

$$\phi_2 = 3(2 - 1) = \boxed{3}$$

$$\phi_3 = 5(3 - 4) = \boxed{-5}$$

Verification:

$$-2 + 3 - 5 = \boxed{-4}$$

$$f(x) - f(z) = 29 - 33 = \boxed{-4}$$

## Experiment 4 – PDP-Based Feature Importance

Given model:

$$f(x) = 0.01(Age) + 0.0001(Income) + Category$$

Dataset:

| Age | Income | Category |
|---|---|---|
| 20 | 30000 | A |
| 30 | 40000 | B |
| 40 | 50000 | A |
| 50 | 60000 | C |

Category values: A = 0.1, B = 0.2, C = 0.3.

**1. PDP for Age**

$$PDP = [4.875, 4.975, 5.075, 5.175]$$

Mean:

$$P\bar{D}P = 5.025$$

Importance:

$$I = \sqrt{\frac{0.0225 + 0.0025 + 0.0025 + 0.0225}{3}}$$

$$\boxed{I(Age) = 0.1291}$$

**2. PDP for Income**

$$PDP = [3.525, 4.525, 5.525, 6.525]$$

Mean:

$$P\bar{D}P = 5.025$$

Importance:

$$I = \sqrt{\frac{2.25 + 0.25 + 0.25 + 2.25}{3}}$$

$$\boxed{I(Income) = 1.2910}$$

**3. PDP for Category**

$$PDP(A) = 4.95$$

$$PDP(B) = 5.05$$

$$PDP(C) = 5.15$$

Importance:

$$I = \frac{5.15 - 4.95}{4}$$

$$\boxed{I(Category) = 0.05}$$

**Ranking**

1. Income = 1.2910
2. Age = 0.1291
3. Category = 0.0500

## Experiment 5 – ICE Plot

Given:

$$f(x_1, x_2) = 3.5 + 3x_1 + 4x_2$$

Dataset:

$$X = [(5, 2), (6, 3), (7, 4)]$$

Grid:

$$x_1 = [5, 6, 7]$$

**Sample 1:** $x_2 = 2$

$$f(5, 2) = 3.5 + 15 + 8 = \boxed{26.5}$$

$$f(6, 2) = 3.5 + 18 + 8 = \boxed{29.5}$$

$$f(7, 2) = 3.5 + 21 + 8 = \boxed{32.5}$$

**Sample 2:** $x_2 = 3$

$$f(5, 3) = \boxed{30.5}$$

$$f(6, 3) = \boxed{33.5}$$

$$f(7, 3) = \boxed{36.5}$$

**Sample 3:** $x_2 = 4$

$$f(5, 4) = \boxed{34.5}$$

$$f(6, 4) = \boxed{37.5}$$

$$f(7, 4) = \boxed{40.5}$$

**PDP**

$$PDP(5) = \frac{26.5 + 30.5 + 34.5}{3} = \boxed{30.5}$$

$$PDP(6) = \frac{29.5 + 33.5 + 37.5}{3} = \boxed{33.5}$$

$$PDP(7) = \frac{32.5 + 36.5 + 40.5}{3} = \boxed{36.5}$$

## Experiment 6 – XRAI

Attribution matrix

$$A = \begin{bmatrix} 0.1 & 0.2 & 0.1 & 0.0 \\ 0.2 & 0.3 & 0.2 & 0.1 \\ 0.0 & 0.1 & 0.8 & 0.7 \\ 0.1 & 0.2 & 0.9 & 0.8 \end{bmatrix}$$

**Region R1**

$$R1 = \frac{0.1 + 0.2 + 0.2 + 0.3}{4} = \boxed{0.20}$$

**Region R2**

$$R2 = \frac{0.1 + 0.0 + 0.2 + 0.1}{4} = \boxed{0.10}$$

**Region R3**

$$R3 = \frac{0.0 + 0.1 + 0.1 + 0.2}{4} = \boxed{0.10}$$

**Region R4**

$$R4 = \frac{0.8 + 0.7 + 0.9 + 0.8}{4} = \boxed{0.80}$$

**Ranking**

1. R4 = 0.80
2. R1 = 0.20
3. R2 = 0.10
4. R3 = 0.10

$$\boxed{\text{Most Important Region} = \text{R4}}$$

## Experiment 7 – Grad-CAM

Given two feature maps:

$$A_1 = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} \qquad A_2 = \begin{bmatrix} 2 & 1 \\ 4 & 3 \end{bmatrix}$$

Gradients:

$$G_1 = \begin{bmatrix} 0.1 & 0.2 \\ 0.3 & 0.4 \end{bmatrix} \qquad G_2 = \begin{bmatrix} 0.2 & 0.1 \\ 0.2 & 0.3 \end{bmatrix}$$

**1. Calculate weights**

Number of spatial locations:

$$Z = 2 \times 2 = 4$$

For feature map 1:

$$\alpha_1 = \frac{0.1 + 0.2 + 0.3 + 0.4}{4}$$

$$\boxed{\alpha_1 = 0.25}$$

For feature map 2:

$$\alpha_2 = \frac{0.2 + 0.1 + 0.2 + 0.3}{4}$$

$$\boxed{\alpha_2 = 0.20}$$

**2. Multiply feature maps by weights**

$$0.25 A_1 = \begin{bmatrix} 0.25 & 0.50 \\ 0.75 & 1.00 \end{bmatrix}$$

$$0.20 A_2 = \begin{bmatrix} 0.40 & 0.20 \\ 0.80 & 0.60 \end{bmatrix}$$

**3. Add weighted feature maps**

$$M = \begin{bmatrix} 0.25 + 0.40 & 0.50 + 0.20 \\ 0.75 + 0.80 & 1.00 + 0.60 \end{bmatrix}$$

$$M = \begin{bmatrix} 0.65 & 0.70 \\ 1.55 & 1.60 \end{bmatrix}$$

**4. Apply ReLU**

$$ReLU(x) = \max(0, x)$$

$$\boxed{L^{Grad-CAM} = \begin{bmatrix} 0.65 & 0.70 \\ 1.55 & 1.60 \end{bmatrix}}$$

**5. Normalize heatmap**

Maximum value = 1.60.

$$L_{norm} = \frac{L}{1.60}$$

$$\boxed{\begin{bmatrix} 0.406 & 0.438 \\ 0.969 & 1.000 \end{bmatrix}}$$

## Experiment 8 – LIME Using Weighted Least Squares

**1. Given data**

Original observation:

$$x = [1, 1]$$

Perturbed observations:

$$z_1 = [1, 1], \quad z_2 = [1, 0]$$

$$z_3 = [0, 1], \quad z_4 = [0, 0]$$

Black-box predictions:

$$y = [0.90, 0.70, 0.60, 0.20]$$

Kernel width:

$$\sigma = 1$$

**2. Calculate Euclidean distances**

$$d(x, z) = \sqrt{(x_1 - z_1)^2 + (x_2 - z_2)^2}$$

$$d_1 = \sqrt{(1-1)^2 + (1-1)^2} = \boxed{0}$$

$$d_2 = \sqrt{(1-1)^2 + (1-0)^2} = \boxed{1}$$

$$d_3 = \sqrt{(1-0)^2 + (1-1)^2} = \boxed{1}$$

**Most important region:** Bottom-right pixel (value = 1.000).

$$d_4 = \sqrt{(1-0)^2 + (1-0)^2} = \boxed{1.414}$$

### 3. Calculate similarity weights

Formula:

$$w_i = e^{-d_i^2/\sigma^2}$$

$$w_1 = e^0 = \boxed{1}$$

$$w_2 = e^{-1} = \boxed{0.368}$$

$$w_3 = e^{-1} = \boxed{0.368}$$

$$w_4 = e^{-2} = \boxed{0.135}$$

### 4. Construct design matrix X

$$X = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 0 \\ 1 & 0 & 1 \\ 1 & 0 & 0 \end{bmatrix}$$

### 5. Construct weight matrix W

$$W = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0.368 & 0 & 0 \\ 0 & 0 & 0.368 & 0 \\ 0 & 0 & 0 & 0.135 \end{bmatrix}$$

### 6. Construct prediction matrix y

$$y = \begin{bmatrix} 0.90 \\ 0.70 \\ 0.60 \\ 0.20 \end{bmatrix}$$

### 7. Calculate $X^T W X$

$$X^T W X = \begin{bmatrix} 1.871 & 1.368 & 1.368 \\ 1.368 & 1.368 & 0.368 \\ 1.368 & 0.368 & 1.368 \end{bmatrix}$$

### 8. Calculate $X^T W y$

$$X^T W y = \begin{bmatrix} 1.600 \\ 1.158 \\ 1.079 \end{bmatrix}$$

### 9. Calculate coefficients

Formula:

$$\beta = (X^T W X)^{-1} X^T W y$$

Solving the system:

$$\begin{bmatrix} 1.871 & 1.368 & 1.368 \\ 1.368 & 1.368 & 0.368 \\ 1.368 & 0.368 & 1.368 \end{bmatrix} \begin{bmatrix} \beta_0 \\ \beta_1 \\ \beta_2 \end{bmatrix} = \begin{bmatrix} 1.600 \\ 1.158 \\ 1.079 \end{bmatrix}$$

$$\boxed{\beta_0 = 0.5712}$$

↓

$$\boxed{\beta_1 = 0.2337}$$

$$\boxed{\beta_2 = 0.1547}$$

### 10. Final interpretation

| Feature | Coefficient | Contribution |
|---|---|---|
| Blood Pressure | +0.2337 | Positive |
| Cholesterol | +0.1547 | Positive |

$$\boxed{\beta_1 > \beta_2 > 0}$$

Blood pressure has a larger positive local contribution than cholesterol.

### 11. Test Case – Negative Feature Contribution

Given:

| X1 | X2 | Prediction |
|---|---|---|
| 1 | 1 | 0.80 |
| 1 | 0 | 0.90 |
| 0 | 1 | 0.50 |
| 0 | 0 | 0.40 |

Effect of X1:

$$0.90 - 0.50 = \boxed{+0.40}$$

Effect of X2:

$$0.80 - 0.90 = \boxed{-0.10}$$

Therefore:

↓

$$\boxed{X1 = \text{Positive contribution}}$$

$$\boxed{X2 = \text{Negative contribution}}$$

### 12. Test Case – Different Kernel Widths

$$w = e^{-d^2/\sigma^2}$$

Distances: $0, 1, 1, 1.414$.

| Distance | $\sigma = 0.5$ | $\sigma = 1$ | $\sigma = 2$ |
|---|---|---|---|
| 0 | 1.000 | 1.000 | 1.000 |
| 1 | 0.018 | 0.368 | 0.779 |
| 1 | 0.018 | 0.368 | 0.779 |
| 1.414 | 0.0003 | 0.135 | 0.607 |

## Experiment 9 – CBOW Word Embeddings and Gradient × Input

### Part A – CBOW

**1. Tokenization**

Given sentences:

- the movie was good
- the movie was bad

Tokens:

$$S_1 = [the, movie, was, good]$$

$$S_2 = [the, movie, was, bad]$$

**2. Vocabulary**

$$V = [the, movie, was, good, bad]$$

$$|V| = 5$$

**3. One-hot encoding**

$$the = [1, 0, 0, 0, 0]$$

$$movie = [0, 1, 0, 0, 0]$$

$$was = [0, 0, 1, 0, 0]$$

$$good = [0, 0, 0, 1, 0]$$

$$bad = [0, 0, 0, 0, 1]$$

**4. CBOW training examples**

Context window = 1.

| Context | Target |
|---|---|
| the, was | movie |
| movie, good | was |
| the, was | movie |
| movie, bad | was |

**5. Matrix dimensions**

Vocabulary size = 5.

Embedding dimension = 2.

$$W = 5 \times 2$$

$$W' = 2 \times 5$$

**6. Average context embedding**

Suppose:

$$e_{movie} = [0.2, 0.4]$$

$$e_{was} = [0.6, 0.8]$$

Then:

$$h = \frac{e_{movie} + e_{was}}{2}$$

$$= \frac{[0.2 + 0.6,\ 0.4 + 0.8]}{2}$$

$$\boxed{h = [0.4, 0.6]}$$

**7. Calculate output scores**

Suppose output matrix:

$$W' = \begin{bmatrix} 0.1 & 0.2 & 0.3 & 0.4 & 0.5 \\ 0.2 & 0.1 & 0.2 & 0.3 & 0.4 \end{bmatrix}$$

For each vocabulary word:

$$z_j = h_1 W'_{1j} + h_2 W'_{2j}$$

$$z_1 = (0.4)(0.1) + (0.6)(0.2) = 0.16$$

$$z_2 = (0.4)(0.2) + (0.6)(0.1) = 0.14$$

$$z_3 = (0.4)(0.3) + (0.6)(0.2) = 0.24$$

$$z_4 = (0.4)(0.4) + (0.6)(0.3) = 0.34$$

$$z_5 = (0.4)(0.5) + (0.6)(0.4) = 0.44$$

**8. Softmax**

$$P_j = \frac{e^{z_j}}{\sum e^{z_k}}$$

Approximate probabilities:

$$\boxed{P = [0.181,\ 0.178,\ 0.196,\ 0.217,\ 0.228]}$$

**9. Cross-entropy loss**

Suppose target word is `movie` (index 1).

$$L = -\log(P(movie))$$

$$= -\log(0.178)$$

$$\boxed{L \approx 1.726}$$

**10. Gradient calculation**

Suppose predicted probabilities are:

$$\hat{y} = [0.10, 0.20, 0.15, 0.45, 0.10]$$

Target is `good`:

$$y = [0, 0, 0, 1, 0]$$

Gradient:

$$\frac{\partial L}{\partial z_j} = \hat{y}_j - y_j$$

$$\boxed{[0.10, 0.20, 0.15, -0.55, 0.10]}$$

**11. Output matrix update**

$$W'_{new} = W'_{old} - \eta(h)(\hat{y} - y)$$

For learning rate $\eta = 0.05$, first output weight:

$$W'_{11,new} = 0.1 - (0.05)(0.4)(0.10)$$

$$\boxed{0.098}$$

**12. Embedding matrix update**

Suppose:

$$W'(\hat{y} - y) = \begin{bmatrix} -0.02 \\ -0.12 \end{bmatrix}$$

Then:

$$\frac{\partial L}{\partial h} = \begin{bmatrix} -0.02 \\ -0.12 \end{bmatrix}$$

For context word `movie`:

$$e_{movie,new} = e_{movie} - \frac{0.05}{2}\begin{bmatrix} -0.02 \\ -0.12 \end{bmatrix}$$

$$= [0.2, 0.4] - [-0.0005, -0.003]$$

$$\boxed{e_{movie,new} = [0.2005, 0.403]}$$

*Note: The CBOW code trains for 1,000 epochs with random initialization. Its final embeddings and cosine similarities depend on the training updates; the manual does not specify one fixed final numerical answer.*

**Part B – Gradient × Input**

Given:

$$w = [0.1, 0.2, 0.1, 2, -2]$$

$$b = 0$$

**1. Positive sentence: "the movie was good"**

Input:

$$x = [1, 1, 1, 1, 0]$$

Calculate:

$$z = w^T x + b$$

$$= (0.1)(1) + (0.2)(1) + (0.1)(1) + (2)(1) + (-2)(0)$$

$$\boxed{z = 2.4}$$

Sigmoid:

$$\hat{y} = \frac{1}{1 + e^{-2.4}}$$

$$\boxed{\hat{y} = 0.9168}$$

**2. Calculate sigmoid gradient**

$$\hat{y}(1 - \hat{y})$$

$$= 0.9168(1 - 0.9168)$$

$$\boxed{0.0763}$$

**3. Calculate individual gradients**

$$Gradient_i = 0.0763 w_i$$

| Word | Calculation | Gradient |
|---|---|---|
| the | $0.0763(0.1)$ | 0.00763 |
| movie | $0.0763(0.2)$ | 0.01526 |
| was | $0.0763(0.1)$ | 0.00763 |
| good | $0.0763(2)$ | 0.1526 |
| bad | $0.0763(-2)$ | -0.1526 |

**4. Gradient × Input**

$$A_i = x_i \times Gradient_i$$

| Word | Input | Gradient | Attribution |
|---|---|---|---|
| the | 1 | 0.00763 | 0.00763 |
| movie | 1 | 0.01526 | 0.01526 |
| was | 1 | 0.00763 | 0.00763 |
| good | 1 | 0.1526 | 0.1526 |
| bad | 0 | -0.1526 | 0 |

$$\boxed{\text{Most important word} = \text{good}}$$

**5. Negative sentence: "the movie was bad"**

Input:

$$x = [1, 1, 1, 0, 1]$$

Calculate:

$$z = (0.1) + (0.2) + (0.1) + (2)(0) + (-2)(1)$$

$$\boxed{z = -1.6}$$

Sigmoid:

$$\hat{y} = \frac{1}{1 + e^{1.6}}$$

$$\boxed{\hat{y} = 0.1680}$$

Sigmoid gradient:

$$0.1680(1 - 0.1680)$$

$$\boxed{0.1398}$$

**6. Gradient × Input**

| Word | Input | Gradient | Attribution |
|---|---|---|---|
| the | 1 | 0.01398 | +0.01398 |
| movie | 1 | 0.02796 | +0.02796 |
| was | 1 | 0.01398 | +0.01398 |
| good | 0 | +0.2796 | 0 |
| bad | 1 | -0.2796 | -0.2796 |

$$\boxed{\text{Most important word} = \text{bad}}$$

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**Experiment 10 – Layer-wise Relevance Propagation (LRP)**

Given weights:

$$w_{11} = 1, \quad w_{21} = 0$$

$$w_{12} = 0, \quad w_{22} = 2$$

$$v_1 = 1, \quad v_2 = 1$$

**1. Forward propagation**

$$h_1 = ReLU(x_1 w_{11} + x_2 w_{21})$$

$$h_2 = ReLU(x_1 w_{12} + x_2 w_{22})$$

$$s = h_1 v_1 + h_2 v_2$$

$$P = \frac{1}{1 + e^{-s}}$$

**2. Relevance propagation**

Output relevance:

$$R_{output} = s$$

Hidden relevance:

$$R_{h1} = \frac{h_1 v_1}{h_1 v_1 + h_2 v_2} R_{output}$$

$$R_{h2} = \frac{h_2 v_2}{h_1 v_1 + h_2 v_2} R_{output}$$

Input relevance:

$$R_{x1} = \frac{x_1 w_{11}}{d_1} R_{h1} + \frac{x_1 w_{12}}{d_2} R_{h2}$$

$$R_{x2} = \frac{x_2 w_{21}}{d_1} R_{h1} + \frac{x_2 w_{22}}{d_2} R_{h2}$$

where:

$$d_1 = x_1 w_{11} + x_2 w_{21}$$

$$d_2 = x_1 w_{12} + x_2 w_{22}$$

**Test Case 1:** $x_1 = 1,\ x_2 = 1$

Hidden activations:

$$h_1 = ReLU(1(1) + 1(0)) = \boxed{1}$$

$$h_2 = ReLU(1(0) + 1(2)) = \boxed{2}$$

Output:

$$s = 1(1) + 2(1) = \boxed{3}$$

Probability:

$$P = \frac{1}{1 + e^{-3}} = \boxed{0.9526}$$

Output relevance:

$$R_{output} = 3$$

Hidden relevance:

$$R_{h1} = \frac{1}{3}(3) = \boxed{1}$$

$$R_{h2} = \frac{2}{3}(3) = \boxed{2}$$

Input relevance:

$$R_{x1} = \frac{1}{1}(1) + 0 = \boxed{1}$$

$$R_{x2} = 0 + \frac{2}{2}(2) = \boxed{2}$$

Conservation:

$$1 + 2 = \boxed{3}$$

**Test Case 2:** $x_1 = 2,\ x_2 = 1$

Hidden activations:

$$h_1 = 2, \quad h_2 = 2$$

Output:

$$s = 2 + 2 = \boxed{4}$$

Probability:

$$P = \frac{1}{1 + e^{-4}} = \boxed{0.9820}$$

Hidden relevance:

$$R_{h1} = \frac{2}{4}(4) = \boxed{2}$$

$$R_{h2} = \frac{2}{2}(4) = \boxed{2}$$

Input relevance:

$$R_{x1} = \boxed{2}$$

$$R_{x2} = \boxed{2}$$

Conservation:

$$2 + 2 = \boxed{4}$$

**Test Case 3:** $x_1 = 0,\ x_2 = 1$

Hidden activations:

$$h_1 = 0, \quad h_2 = 2$$

Output:

$$s = 0 + 2 = \boxed{2}$$

Probability:

$$P = \frac{1}{1 + e^{-2}} = \boxed{0.8808}$$

Hidden relevance:

$$R_{h1} = \boxed{0}$$

$$R_{h2} = \boxed{2}$$

Input relevance:

$$R_{x1} = \boxed{0}$$

$$R_{x2} = \boxed{2}$$

Conservation:

$$0 + 2 = \boxed{2}$$

**Test Case 4:** $x_1 = 1,\ x_2 = 0$

Hidden activations:

$$h_1 = 1, \quad h_2 = 0$$

Output:

$$s = 1 + 0 = \boxed{1}$$

Probability:

$$P = \frac{1}{1 + e^{-1}} = \boxed{0.7311}$$

Hidden relevance:

$$R_{h1} = \boxed{1}$$

$$R_{h2} = \boxed{0}$$

Input relevance:

$$R_{x1} = \boxed{1}$$

$$R_{x2} = \boxed{0}$$

Conservation:

$$1 + 0 = \boxed{1}$$

**Test Case 5:** $x_1 = 0,\ x_2 = 0$

Hidden activations:

$$h_1 = 0, \quad h_2 = 0$$

Output:

$$s = 0$$

Probability:

$$P = \frac{1}{1 + e^{0}} = \boxed{0.5}$$

Since the denominator is zero, the program assigns all relevance values zero.

$$\boxed{R_{h1} = R_{h2} = R_{x1} = R_{x2} = 0}$$

**Final answers**

| Test case | Input | Hidden activations | Output logit | Probability | Input relevance |
|---|---|---|---|---|---|
| 1 | (1,1) | (1,2) | 3 | 0.9526 | (1,2) |
| 2 | (2,1) | (2,2) | 4 | 0.9820 | (2,2) |
| 3 | (0,1) | (0,2) | 2 | 0.8808 | (0,2) |
| 4 | (1,0) | (1,0) | 1 | 0.7311 | (1,0) |
| 5 | (0,0) | (0,0) | 0 | 0.5000 | (0,0) |

For the first four cases:

$$\boxed{R_{x1} + R_{x2} = R_{output}}$$